

COVID VACCINATION RATES IN THE U.S.

A comparison between Portugal's Vaccination Rate
and the United States' Vaccination Rate
When will the U.S. Population Be 90% Vaccinated?

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RESEARCH QUESTION

When will 90% of the US population be fully vaccinated?

- Fully Vaccinated Meaning
 - Receives full dosage of a particular COVID-19 Vaccine
 - Pfizer
 - 2 doses
 - Moderna
 - 2 doses
 - Johnson & Johnson
 - 1 dose



WHY IS THIS IMPORTANT?



- **Covid-19 Cases Statistics**

- As of November 18, 2021, there have been about 256 million Covid-19 Cases worldwide (Statista)
 - 48 million people have contracted Covid-19 alone in the US
- Around 5.1 million have died worldwide from Covid-19 (Statista)
 - 783,565 have died in the U.S. from Covid-19
- Currently 69% of the U.S. population is vaccinated with only 59% being fully vaccinated (The New York Times)

- **Covid-19 Importance**

- Direct Impact on Our Community
- Even those that recovered still can suffer from lasting health issues due to Covid-19 implications
 - long-term breathing problems
 - heart complications
 - chronic kidney impairment
- Adopting 90% full vaccination in the U.S. would help minimize death from Covid-19

SOURCES OF BIAS

- Individuals that cannot get the vaccine
 - Allergic to certain vaccine ingredients
 - People with certain medical conditions
- People affected by homelessness
 - According to the National Academy for State Health Policy, "...at least 20 states don't include people living in homeless shelters in their vaccine distribution plans..." (Pew Trusts)
- Accessibility/Proximity to Nearest Provider
- Issues with Processing the Data
 - Incomplete/missing data for certain days



ALGORITHM

MACHINE LEARNING ALGORITHM USED



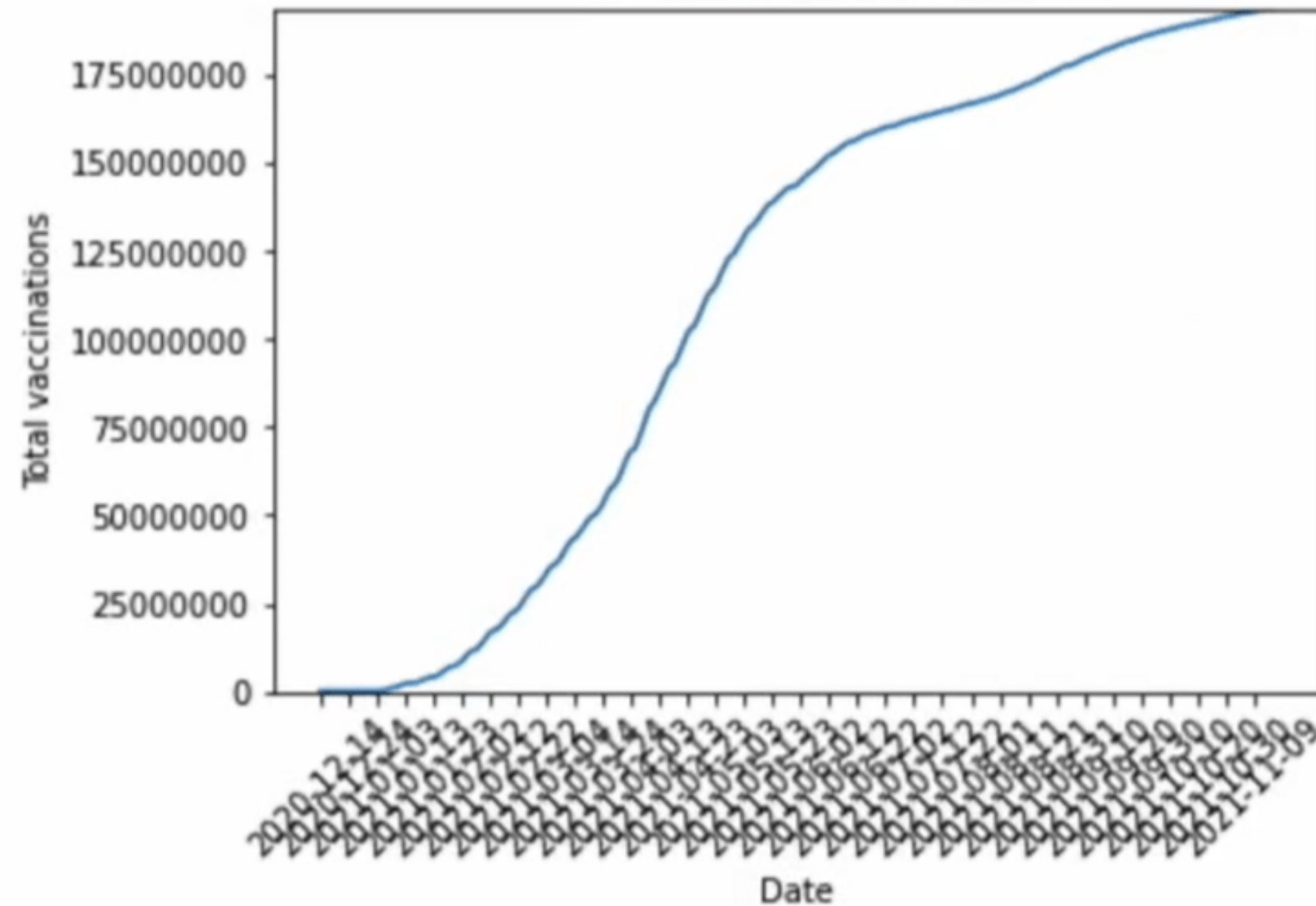
- Linear Regression Model

- Supervised Machine Learning Model
- "...assumes a linear relationship between variables." (Jidge, Abhay.)
- "One variable is considered to be an explanatory variable, and the other is considered to be a dependent variable. (Yale.)"

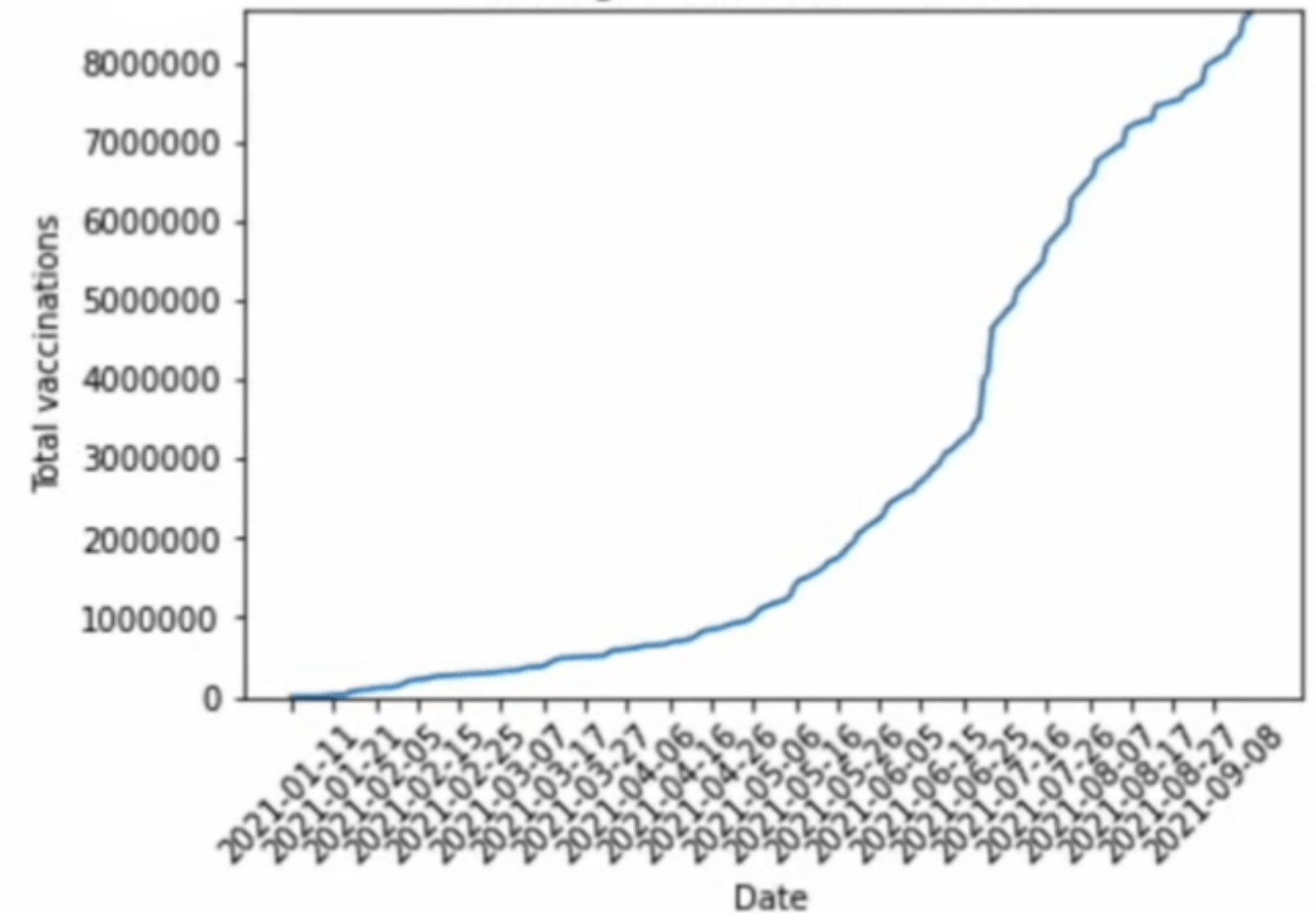
**DATA
VISUALIZATIONS,
MODEL DEMO,
& MODEL
EVALUATION**

DATA VISUALIZATION SUMMARY

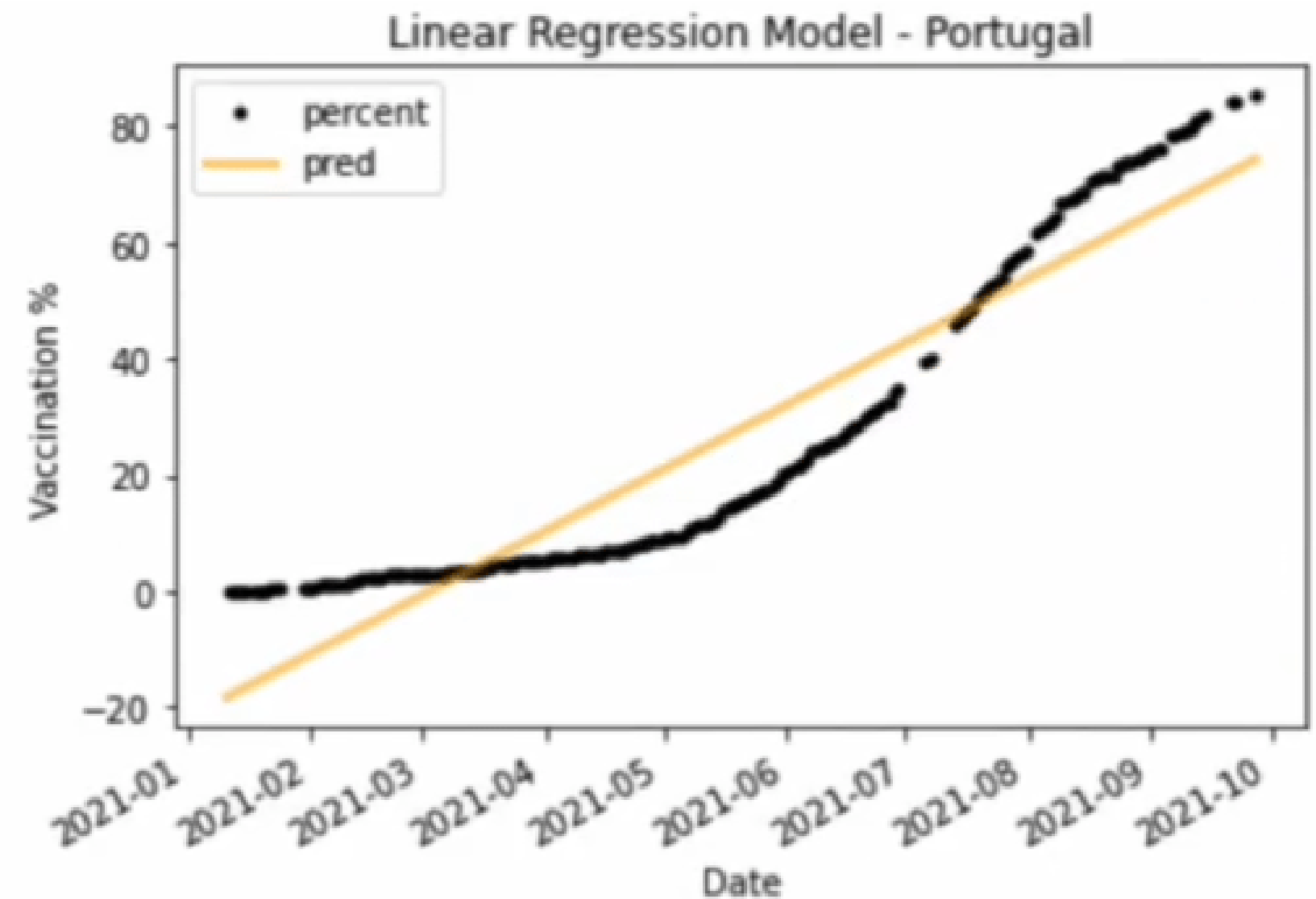
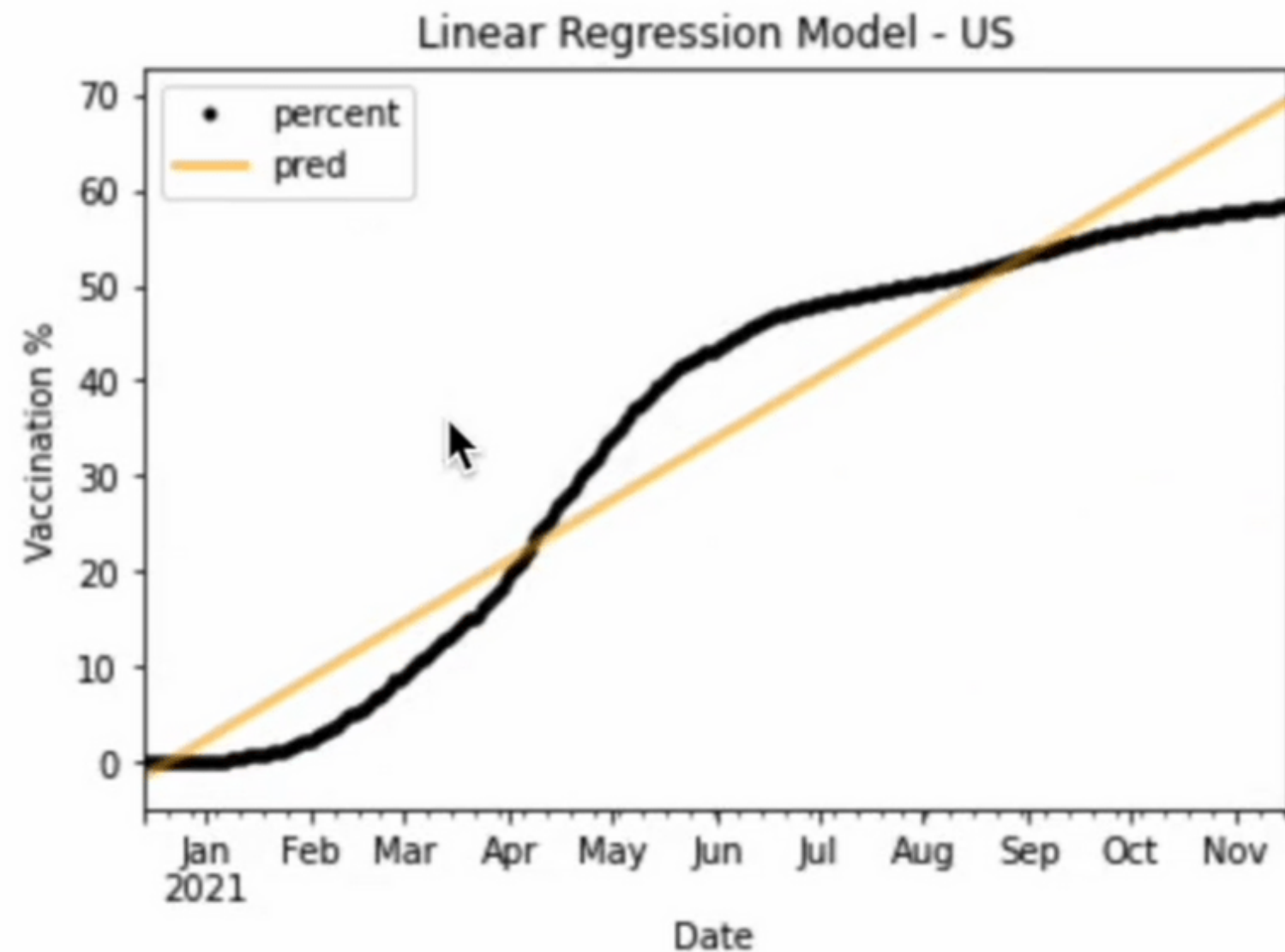
US Vaccinations in Total



Portugal Vaccinations in Total



DATA VISUALIZATION SUMMARY



MODEL DEMO & EVALUATION

The image shows a desktop environment with a Windows-style taskbar on the left. The taskbar contains several icons: a folder named 'Misc', a PDF file 'Ekdev Rajkitkul Resume', a PDF file 'Ekdev Rajkitkul HS Transcript.pdf', a PDF file 'Ekdev Rajkitkul - Resume', a PDF file 'Ekdev Rajkitkul Resumé', a folder named 'Prewrite', and a folder named 'latex'. The main window is a web browser displaying the Kaggle dataset page for 'COVID-19 World Vaccination Progress' by Gabriel Preda. The browser's address bar shows the URL 'kaggle.com/gpreda/covid-world-vaccination-progress'. The Kaggle page features a dark header with the title 'COVID-19 World Vaccination Progress' and a subtitle 'Daily and Total Vaccination for COVID-19 in the World from Our World in Data'. Below the header, there is a 'Dataset' section with a 'Download (13 MB)' button. The page also displays 'Usability 10.0', 'License CC0: Public Domain', and 'Tags health, biology, covid19, public health'. The 'Description' section includes a 'Context' paragraph and a 'Content' section. The 'Context' paragraph states: 'Data is collected daily from Our World in Data GitHub repository for covid-19, merged and uploaded. Country level vaccination data is gathered and assembled in one single file. Then, this data file is merged with locations data file to include vaccination sources information. A second file manufacturers information, is included.' The 'Content' section states: 'The data (country vaccinations) contains the following information:'. The 'Recently Viewed' section on the left sidebar lists several datasets, including 'COVID-19 World Vaccination Progress'.

Desktop Environment:

- Misc
- Ekdev Rajkitkul Resume
- Ekdev Rajkitkul HS Transcript.pdf
- Ekdev Rajkitkul - Resume
- Ekdev Rajkitkul Resumé
- Prewrite
- latex

Kaggle Dataset Page:

URL: kaggle.com/gpreda/covid-world-vaccination-progress

Dataset: COVID-19 World Vaccination Progress

Subtitle: Daily and Total Vaccination for COVID-19 in the World from Our World in Data

Author: Gabriel Preda • updated 3 days ago (Version 219)

Download (13 MB)

Usability 10.0 License CC0: Public Domain Tags health, biology, covid19, public health

Description

Context

Data is collected daily from [Our World in Data](#) GitHub repository for [covid-19](#), merged and uploaded. Country level vaccination data is gathered and assembled in one single file. Then, this data file is merged with locations data file to include vaccination sources information. A second file manufacturers information, is included.

Content

The data (country vaccinations) contains the following information:

WHAT HAS PORTUGAL DONE TO ACHIEVE SUCH A HIGH VACCINATION PERCENTAGE?



- Portugal was quick to declare immediate action when the World Health Organization (WHO) made the announcements.
 - Public areas were closed off and citizens were told to stay home until conditions are admirable again.
 - During religious holidays, citizens were only allowed to travel within their towns.
- Border was temporarily closed off. Immigrants who recently settled into Portugal were given access to the same goods/services native citizens had.

WHAT CAN WE DO IN THE U.S. INCREASE OUR VACCINATION NUMBERS?

- Social Media Campaign to raise social awareness around the vaccination as well as help boost vaccine security in individuals
- Local Government can enact an outreach program targeted to people within their community such as through their community centers & shelters
 - Send out volunteers to spread awareness to people that are homeless or those that aren't in shelters regarding vaccination

PORTUGAL VS U.S. VACCINATION DATA

- Portugal

- People Fully Vaccinated Against Covid-19: 87.78% (As of Nov. 15, 2021)
- Share of People Only Partly Vaccinated Against COVID-19: 1.26%
- **Total People Vaccinated: 89.04 % (Our World in Data)**

- U.S.

- People Fully Vaccinated Against Covid-19: 57.81%
- Share of People Only Partly Vaccinated Against COVID-19: 10.36%
- **Total People Vaccinated: 68.17% (Our World in Data)**

- Portugal still has mandatory mask usage in public transportation and in service establishments, bars, clubs, restaurants and commercial establishments for workers.

OUR GITHUB REPOSITORY



- Link to Our GitHub Repository
 - https://github.com/adanRivas/a_i4all_covid19_vax



CONSIDERING THE IMPACT OF OUR WORK



- Positive Impacts

- Gives hope to our community that they will be more protected against COVID-19
- Gives hope to those that are relying on herd immunity
- According to Dr. Ashley Drews, "It's still unclear exactly how many people will need to be vaccinated in order to achieve herd immunity to COVID-19, but experts estimate that it will take at least 70% of the population — with some estimates ranging as high as 90%," (Houston Methodist)

- Negative Impacts

- Mutations of the virus make herd immunity difficult
- We aren't certain how long vaccine immunity lasts
 - "The natural immunity acquired after infection seems to last for several months, and the expectation is that the immunity conferred via vaccination will last even longer. But we don't know exactly how long yet," (Houston Methodist)
- Some people are still apprehensive of getting the vaccine which can delay our progress and throw off predictions
 - May cause panic and anxiety in people

THE ESSENTIAL QUESTION

How Can the Process of Creating AI/ML
Solutions Amplify or Mitigate Bias?

HOW CAN THE PROCESS OF CREATING AI/ML SOLUTIONS AMPLIFY OR MITIGATE BIAS?

By creating a prediction model, the project will be capable of determining how many people are fully vaccinated against COVID-19 through the use of actual data. Consequently, this allows for the project to calculate and output an accurate prediction over how many vaccinated people there are in the world.

However, the data can be easily affected by factors such as people having reasons not to take vaccines, thus it is best to approach the data with caution.



NEXT STEPS

How Will We Continue to Work on Our
Project as a Team Beyond Today?

NEXT STEPS

- Test out different models to try to get a better fit on our data
 - We will experiment with polynomial regression and/or logistic regression to see which one of these models gives us the best fit of our data
 - This will help us get a more accurate reading of when we can possibly expect the US to have 90% fully vaccinated population.
- Finding ways to mitigate bias in our data from those that cannot get the vaccine due to medical issues such as allergies to ingredients in vaccines and other medical conditions

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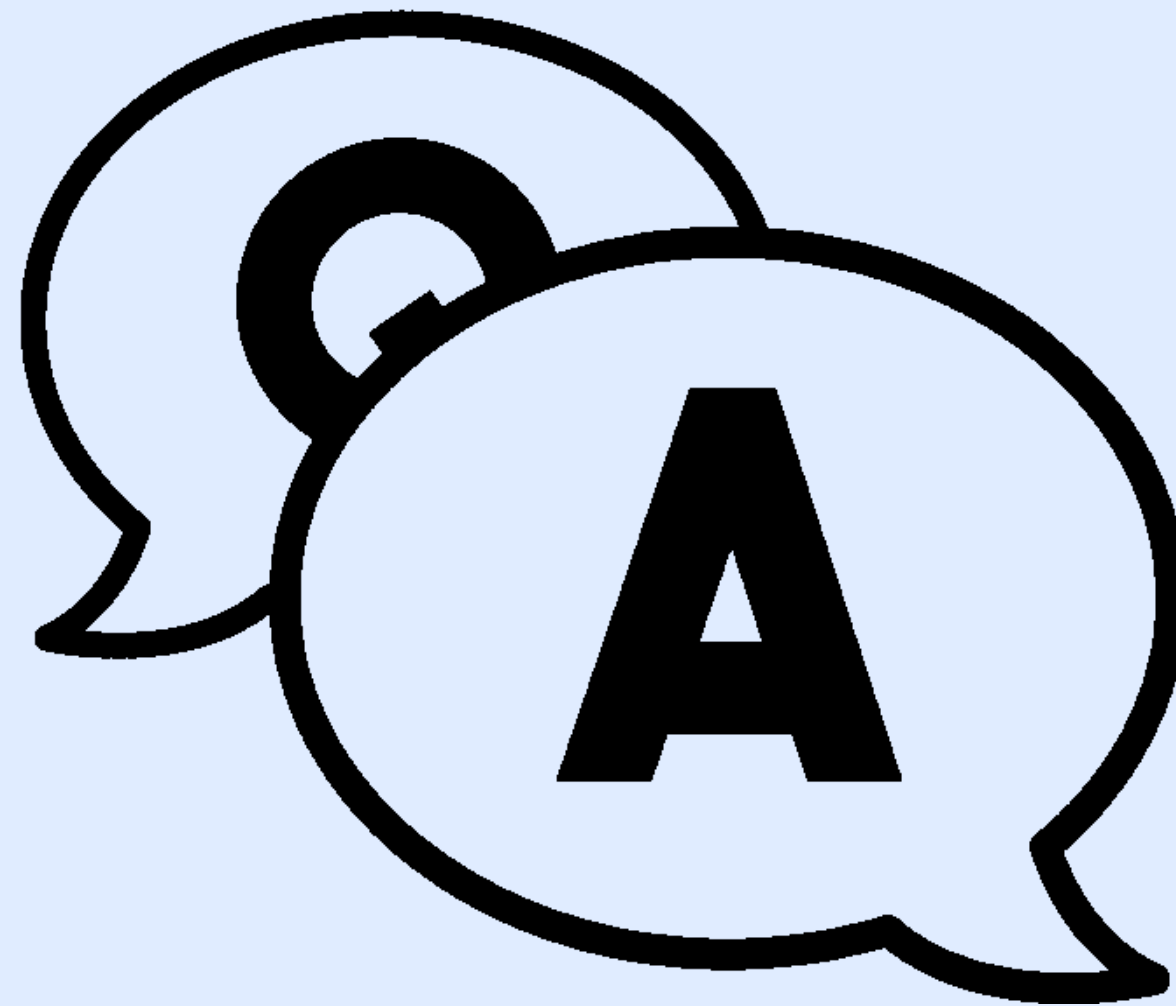
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QUESTIONS?



THANK YOU!

